

Lecture 2: One Parameter Models

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Announcements

- Reading: Chapter 2 (these lecture notes)
- Chapter 3 (next lecture notes)

BDA3

$$P(\theta|y) \neq \frac{P(y|\theta)P(\theta)}{P(y)}$$

posterior

$L(\theta)$
DGP

$P(y)$

prior

Cromwell's Rule

The use of priors placing a probability of 0 or 1 on events should be avoided except where those events are excluded by logical impossibility.

I beseech you, in the bowels of Christ, think it possible that you may be mistaken.

— Oliver Cromwell

If a prior places probabilities of 0 or 1 on an event, then no amount of data can update that prior.

Cromwell's Rule

Leave a little probability for the moon being made of green cheese; it can be as small as 1 in a million, but have it there since otherwise an army of astronauts returning with samples of the said cheese will leave you unmoved.

— Dennis Lindley (1991)

If $p(\theta = a) = 0$ for a value of a , then the posterior distribution is always zero, regardless of what the data says

$$p(\theta = a|y) \propto p(y|\theta = a) \underbrace{p(\theta = a)}_{0} = 0$$

$\mathcal{D}$ $L(\theta) \times 0$

$$Y \sim \text{Bin}(n, \theta), \quad P(\theta) > 0 \quad \forall \theta \in [0, 1]$$

$$Y \sim \text{Pois}(\lambda), \quad P(\lambda) > 0 \quad \forall \lambda \in \mathbb{R}^+$$

$$Y \sim N(\mu, \sigma^2) \quad P(\mu) > 0 \quad \forall \mu \in \mathbb{R}$$

Example: estimating shooting skill in basketball

- On November 18, 2017, an NBA basketball player, Robert Covington, had made 49 out of 100 three point shot attempts.
- At that time, his three point field goal percentage, 0.49, was the best in the league and would have ranked in the top ten all time
- How can we estimate his true shooting skill?
 - Think of “true shooting skill” as the fraction he would make if he took infinitely many shots

$$Y \sim \text{Bin}(100, \theta)$$

(R.C.
Made
3 pt shots)

- Identically distributed shots.
- Independence (fatigue, confidence)

Example: estimating shooting skill in basketball

Prior for θ ?

- Assume every shot is independent (reasonable) and identically distributed (less reasonable?)
- Let $Y \sim \text{Bin}(n, \theta)$ where θ corresponds to his true skill
- Frequentist inference tells us that the maximum likelihood estimate is simply $\frac{y}{n} = 49/100 = 0.49$
- What would our estimates be if we use Bayesian inference?
 - If our prior reflects “complete ignorance” about basketball? $\theta \sim \text{Unif}(0, 1)$
 - What if we want to incorporate prior domain knowledge?

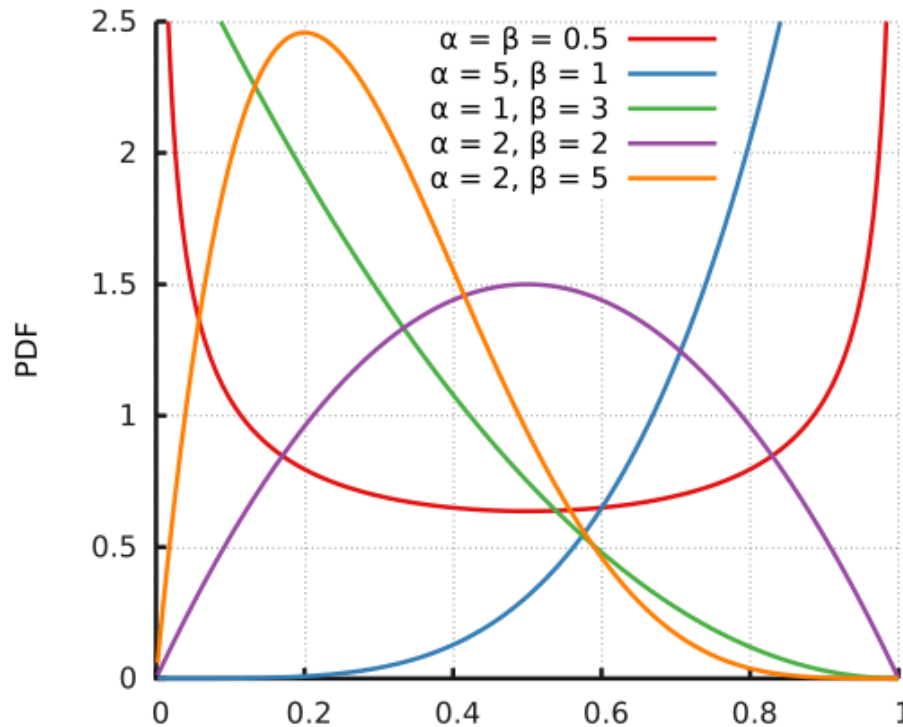
The Binomial Model

- The uniform prior: $p(\theta) = \text{Unif}(0, 1) = \mathbf{1}\{\theta \in [0, 1]\}$
 - A “non-informative” prior
- Posterior: $p(\theta \mid y) \propto \underbrace{\theta^y (1 - \theta)^{n-y}}_{\text{likelihood}} \times \underbrace{\mathbf{1}\{\theta \in [0, 1]\}}_{\text{prior}}$
- The above posterior density is is a density over θ .

$\propto \theta^y (1 - \theta)^{n-y}$

Beta Density: $\text{Beta}(y+1, n-y+1)$

Beta Distributions

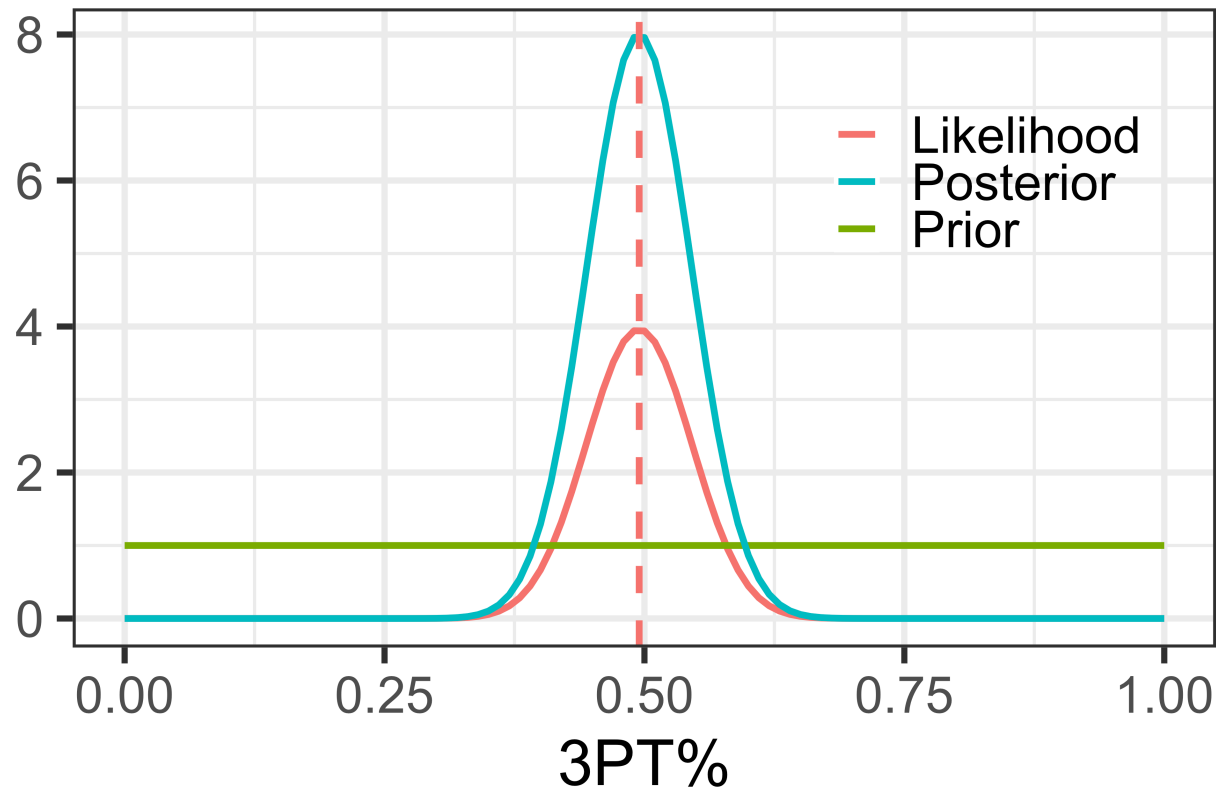


*Normalizing
Constant.*

$$\text{Beta}(\alpha, \beta) = \frac{\Gamma(\alpha + \beta)}{\Gamma(\alpha)\Gamma(\beta)} \theta^{\alpha-1} (1 - \theta)^{\beta-1}$$

Example: estimating shooting skill in basketball

Likelihood, Prior, Posterior



Summarizing Posterior Results

Point Estimates

- $E[\Theta | y] = \int \Theta P(\Theta | y) d\Theta$ "posterior mean"

- $\underset{\Theta}{\operatorname{argmax}} P(\Theta | y)$ "MAP" (maximum a posterior)

- $\operatorname{Var}(\Theta | y)$ "posterior variance"

- $P(\Theta \in R(y)) = 95\%$ 95%
"Credible Interval"
not confidence

Summarizing Posterior Results

- $\text{Beta}(\alpha, \beta) = \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} \theta^{\alpha-1} (1 - \theta)^{\beta-1}$ *Density*
- The mean of a $\text{Beta}(\alpha, \beta)$ distribution r.v. is $\frac{\alpha}{\alpha+\beta}$
- The mode of a $\text{Beta}(\alpha, \beta)$ distributed r.v. is $\frac{\alpha-1}{\alpha+\beta-2}$
- The variance of a $\text{Beta}(\alpha, \beta)$ r.v. is $\frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}$
- In R: dbeta, rbeta, pbeta, qbeta

The Bayesian Advantage

- Particularly effective with small samples
 - With a true shooting skill of 0.35, there is about a 1/4 chance of making 5 or more out of the first 10 shots
- Ability to incorporate real prior knowledge
- Can easily “share information” across related observations

Informative prior distributions

- At that point in November, Covington's three point field goal percentage, 0.49, was the best in the league and would have ranked in the top ten all time
- It seems very unlikely that this level of skill would continue for an entire season of play.
- A uniform prior distribution doesn't reflect our known beliefs. We need to choose a more informative prior distribution

- All other players' Skill

- R.C. Previous Play

Informative prior distributions

- When $p(\theta) \sim U(0, 1)$ then the posterior was a Beta distribution
- Remember: the binomial likelihood is $L(\theta) \propto \theta^y (1 - \theta)^{n-y}$
- Choose a prior with a similar looking form:
 $p(\theta) \propto \theta^{\alpha-1} (1 - \theta)^{\beta-1}$

$$P(\theta/y) \propto \theta^y (1-\theta)^{n-y} \times \theta^{\alpha-1} (1-\theta)^{\beta-1}$$
$$\propto \theta^{y+\alpha-1} (1-\theta)^{n-y+\beta-1}$$

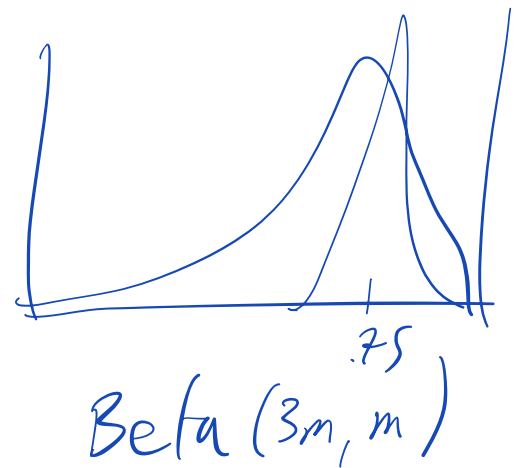
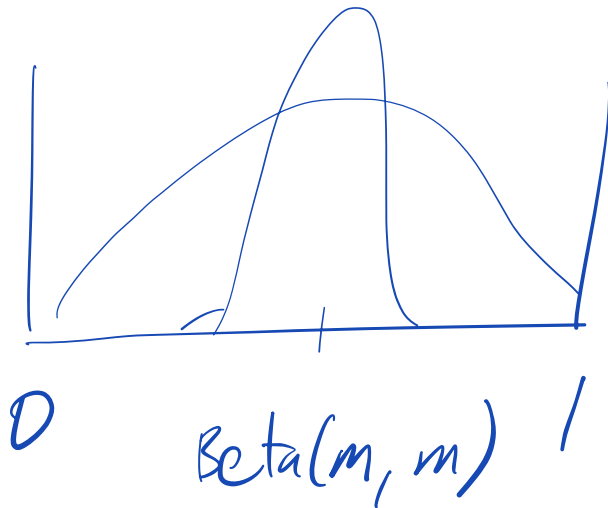
$$\text{Beta}(y+\alpha, n-y+\beta)$$

"pseudo-counts" interpretation.

α : "prior" shots made

β : "prior" missed shots

$$E[\theta] = \frac{\alpha}{\alpha + \beta} \quad (\text{prior mean}), \quad \text{"prior frequency"}$$



Conjugate Prior Distributions

Definition: A class of prior distributions, $\mathcal{P}$ for θ is called *conjugate* for a sampling model $p(Y|\theta)$ if

$$\underline{p(\theta) \in \mathcal{P}} \implies p(\theta|y) \in \underline{\mathcal{P}} \quad \text{Beta} \rightarrow \text{Beta Post.}$$

- The prior distribution and the posterior distribution are in the same family
- Conjugate priors are very convenient because they make calculations easy
- The parameters for conjugate prior distribution have nice interpretations

Pseudo-Counts Interpretation

- Observe y successes, $n - y$ failures
- If $p(\theta) \sim \text{Beta}(\alpha, \beta)$ then
 $p(\theta | y) = \text{Beta}(y + \alpha, n - y + \beta)$
- What is $E[\theta | y]$? (posterior mean)

$$E[\theta | y] = \frac{y + \alpha}{y + \alpha + n - y + \beta} = \frac{y + \alpha}{\alpha + \beta + n}$$

$\alpha + \beta$
 // pseudo
 sample size'

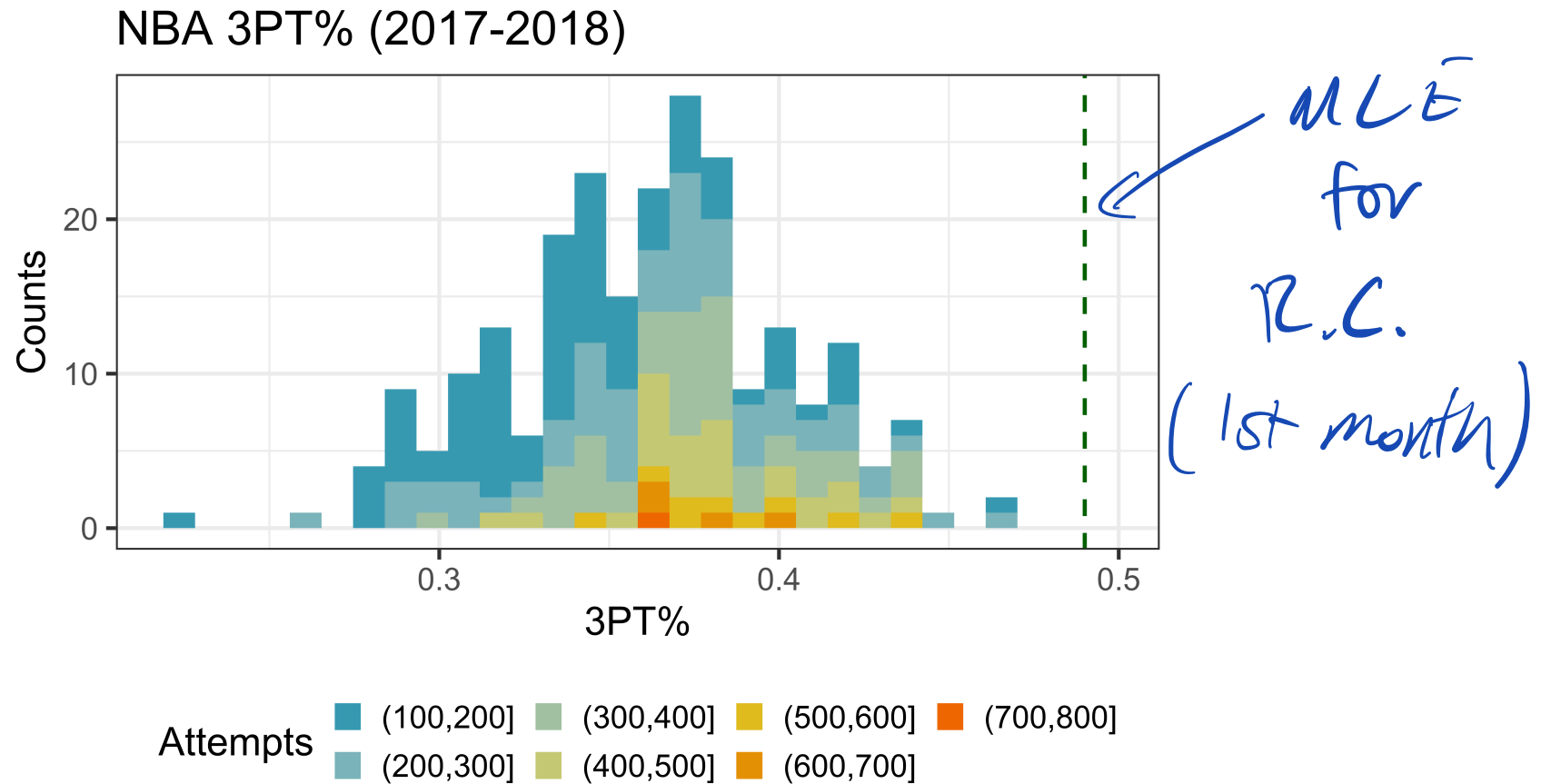
$$= \left(\frac{n}{n + \alpha + \beta} \right) \frac{y}{n} + \frac{\alpha}{\alpha + \beta + n} \left(\frac{\alpha + \beta}{\alpha + \beta} \right)$$

$$= \frac{n}{n + \alpha + \beta} \frac{y}{n} + \frac{\alpha + \beta}{n + \alpha + \beta} \frac{\alpha}{\alpha + \beta} = w \hat{\theta}_{MLE} + (1 - w) \hat{\theta}_{prior}$$

Example: estimating shooting skill in basketball

- On November 18, 2017, an NBA basketball player, Robert Covington, had made 49 out of 100 three point shot attempts.
- At that time, his three point field goal percentage, 0.49, was the best in the league and would have ranked in the 10 ten all time
- Prior knowledge tells us it is unlikely this will continue!
- How can we use Bayesian inference to better estimate his true skill?

Three point shooting in 2017-2018



...

Regression Toward the Mean

What is a reasonable model?

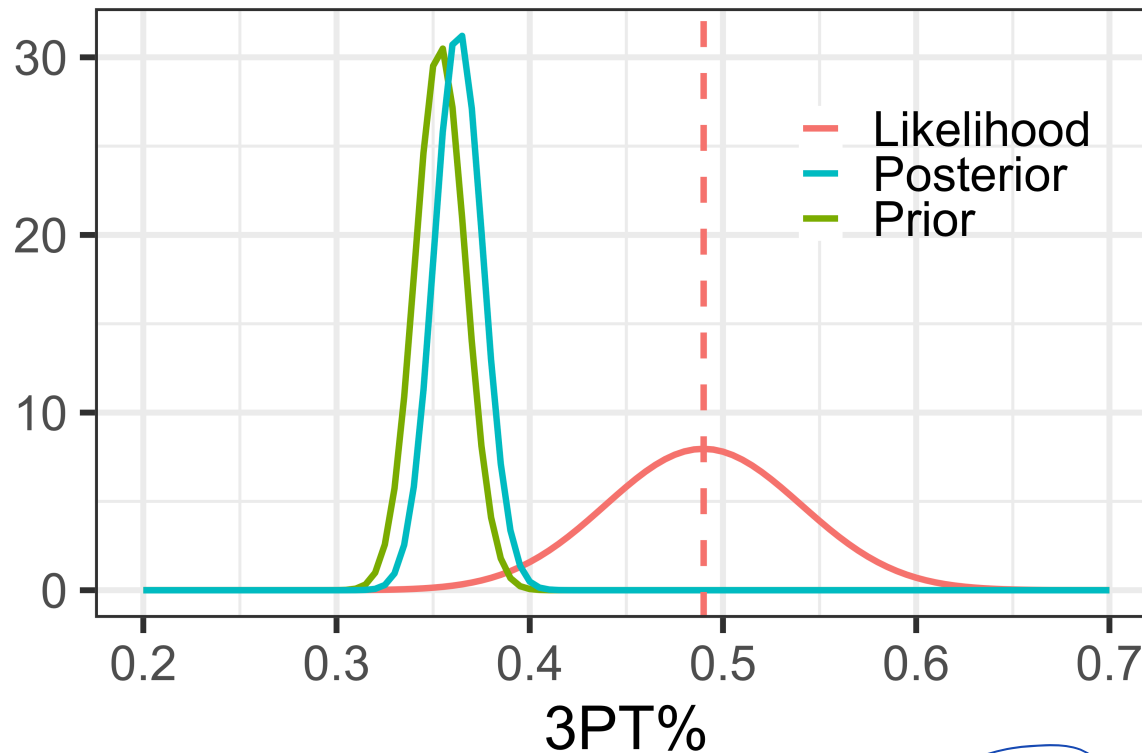
- If we believe that his skill doesn't change much year to year, use past data to inform prior
- In his first 4 seasons combined Robert Covington made a total of 478 out of 1351 three point shots (0.35%, just below average).
- Choose a $\text{Beta}(\alpha, \beta)$ prior (pseudo-count interpretation)

$$P(\theta | y) \sim \text{Beta}(\overset{y}{49} + \overset{\alpha}{478}, \overset{n-y}{51} + \overset{\beta}{873})$$

Robert Covington 2017-2018 estimates

After 100 shots Robert Covington's 3PT% was 0.49

Likelihood, Prior, Posterior

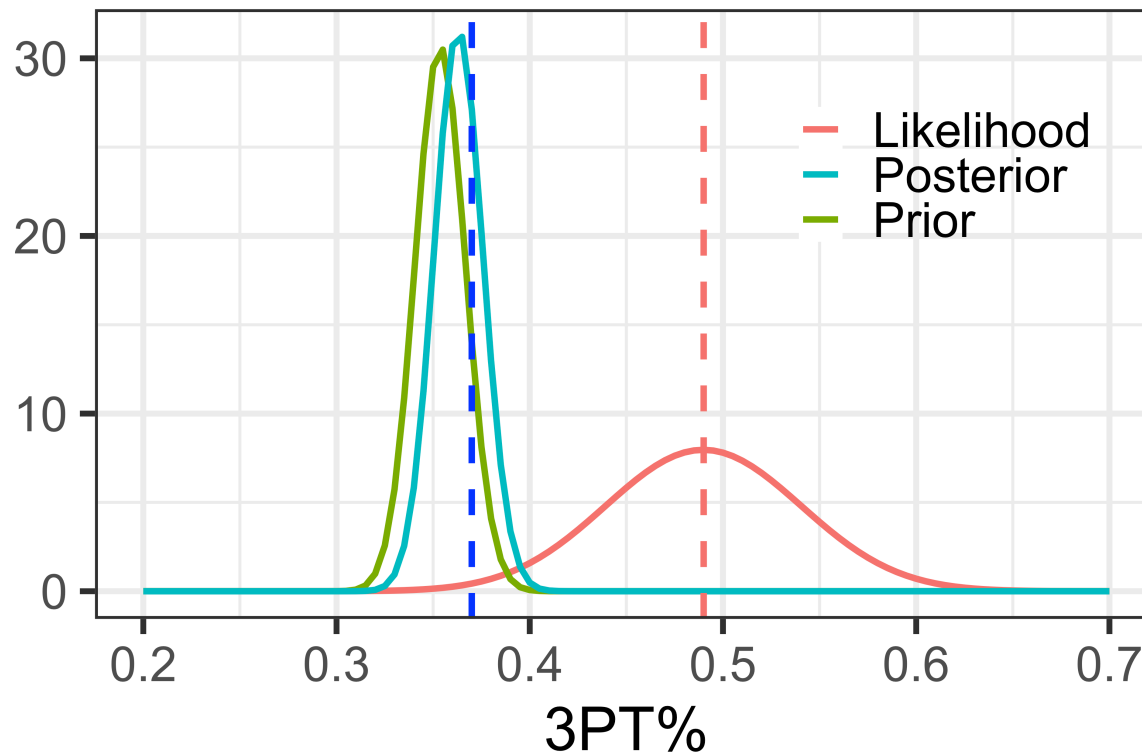


MLE = 0.49, posterior mean = 0.36

How did we do?

Robert Covington's end of season 3PT% was 0.37

Likelihood, Prior, Posterior



MLE = 0.49, posterior mean = 0.36

The Poisson Distribution

- A useful model for count data
- Events occur independently at some rate λ
- Mean = variance = λ .
- Example applications:
 - Epidemiology (disease incidence)
 - Astronomy (e.g. the number of meteorites entering the solar system each year)
 - The number of patients entering the emergency room
 - The number of times a neuron in the brain “fires”

$$Y_i \overset{\text{indep}}{\sim} \text{Pois}(v_i \lambda)$$

$$E[Y_i] = \text{Var}(Y_i) = v_i \lambda$$

$\uparrow$ exposure for unit i .

$\nwarrow$ Expected counts per unit exposure.

$$L(\lambda) \propto \prod_{i=1}^n \frac{e^{-v_i \lambda} (v_i \lambda)^{y_i}}{y_i!} \propto \frac{e^{-(\sum v_i) \lambda} \lambda^{\sum y_i}}{1}$$

$$\underline{P(\lambda)} \propto \lambda^{\alpha-1} e^{-B\lambda} \Rightarrow \text{Gamma}(\alpha, B)$$

$$\underline{P(\lambda | y_1, \dots, y_n)} \propto \lambda^{\alpha + \sum y_i - 1} e^{-\underline{(\sum v_i + B)\lambda}}$$

$$\text{Gamma}(\underline{a} + \sum y_i, \underline{b} + \sum v_i)$$

Poisson model with exposure

- Assume y_i is Poisson with rate λ and exposure ν_i :

$$p(y_i \mid \nu_i \lambda) = (\nu_i \lambda)^{y_i} e^{-\nu_i \lambda} / y_i!$$

- How many cars do we expect to pass an intersection in one hour? How many in two hours?
 - If we model the distribution as Poisson, we expect twice as many in two hours as in one hours.

Conjugate Prior for the Poisson Distribution

The Gamma distribution

We use the shape-rate parameterization of the Gamma

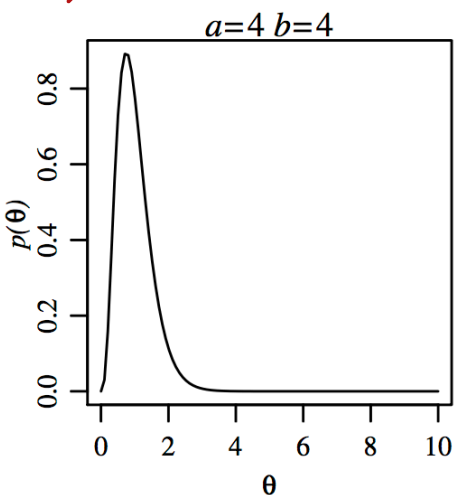
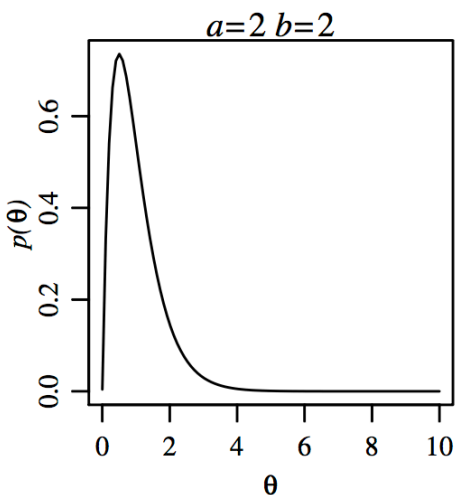
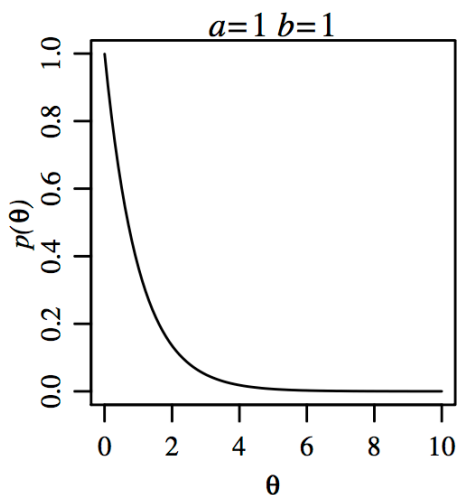
- $\text{Gamma}(a, b) = \frac{b^a}{\Gamma(a)} \lambda^{a-1} e^{-b\lambda}$
- $E[\lambda] = a/b$ and $\text{Var}[\lambda] = a/b^2$
- $\text{mode}[\lambda] = (a - 1)/b$ if $a > 1$, 0 otherwise
- In R: `dgamma`, `rgamma`, `pgamma`, `qgamma`

counts
time

The Gamma distribution

Smaller Variance.

*Mean
1*



*Mean
2*

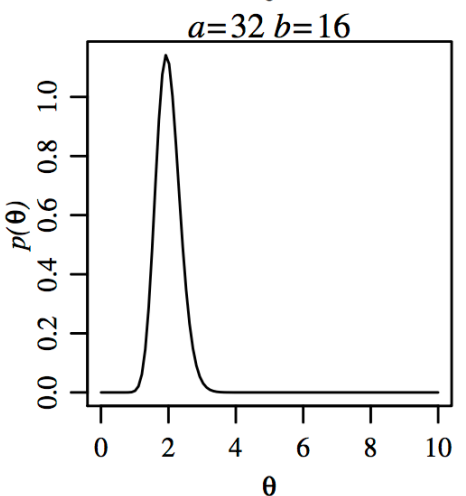
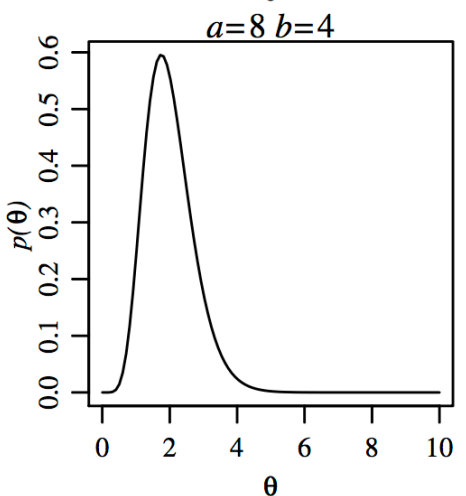
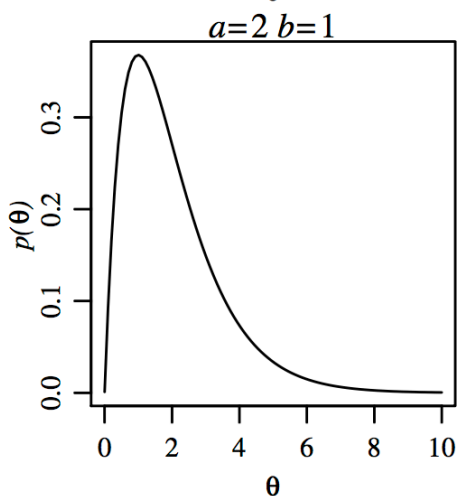


Fig. 3.8. Gamma densities.

The posterior mean

$$\lambda | y_1, \dots, y_n \sim \text{Gam}(\sum y_i + \alpha, \sum v_i + \beta)$$

$$E[\lambda | y_1, \dots, y_n] = \frac{\sum y_i + \alpha}{\sum v_i + \beta}$$

$$= \frac{\sum v_i}{\sum v_i + \beta} \frac{\sum y_i}{\sum v_i} + \frac{\alpha}{\sum v_i + \beta} \left(\frac{\beta}{\beta} \right)$$

$$= \frac{\sum v_i}{\sum v_i + \beta} \frac{\sum y_i}{\sum v_i} + \frac{\beta}{\sum v_i + \beta} \frac{\alpha}{\beta} = w \hat{\lambda}_{MLE} + (1-w) \lambda_{prior}$$

Conjugate Priors and Exponential Families

- An k -dimensional exponential family distribution can be written as $p(y | \theta) = h(y) \exp(\eta(\theta)T(y) - A(\theta))$
- For any k -parameter exponential family there exists a $(k + 1)$ -parameter conjugate prior.

$$L(\theta) \propto \exp \left[\sum_{i=1}^k \underbrace{n_i(\theta)}_{\text{Natural parameter}} \underbrace{T_i(y)}_{\text{k-dim sufficient statistic}} - \underbrace{A(\theta)}_{\text{log Partition.}} \right]$$

parametrized
by α ($k+1$ dim)

$$P(\theta) \propto$$

$$\exp \left[\sum_{i=1}^k n_i(\theta) \alpha_i - \alpha_{k+1} A(\theta) \right]$$

prior
params

$$P(\eta(\theta) / T_1(y), \dots, T_K(y)) \propto \exp \left[\sum_{i=1}^K \eta_i(\theta) (\underbrace{T_i(y) + \alpha_i}_{(1 + \alpha_{K+1}) A(\theta)}) - \right]$$

Uncertainty Quantification

Posterior Credible Intervals

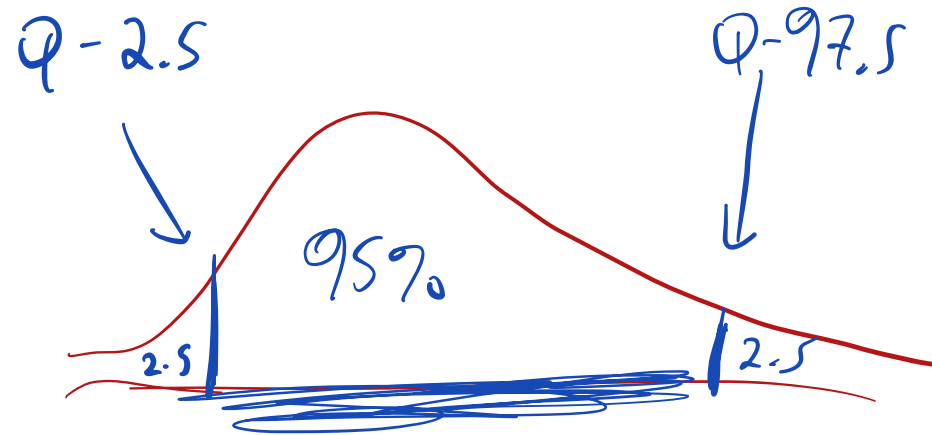
- Frequentist interval: $Pr(l(Y) < \theta < u(Y) \mid \theta) = 0.95$
 - Probability that the interval will cover the true value *before* the data are observed.
 - Interval is random since Y is random.
- Bayesian Interval: $Pr(l(y) < \theta < u(y) \mid Y = y) = 0.95$
 - Information about the the true value of θ *after* observeing $Y = y$.
 - θ is random (because we include a prior), y is observed so interval is non-random.

Posterior Credible Intervals (Quantile-based)

- The easiest way to obtain a credible interval is to use the quantiles of the posterior distribution.

If we want $100 \times (1 - \alpha)$ interval, we find numbers $\theta_{\alpha/2}$ and $\theta_{1-\alpha/2}$ such that:

1. $p(\theta < \theta_{\alpha/2} \mid Y = y) = \alpha/2$
2. $p(\theta > \theta_{1-\alpha/2} \mid Y = y) = \alpha/2$



$$p(\theta \in [\theta_{\alpha/2}, \theta_{1-\alpha/2}] \mid Y = y) = 1 - \alpha$$

Interval for shooting skill in basketball

- The posterior distribution for Covington's shooting percentage is a

$$\text{Beta}(49 + 478, 50 + 873) = \text{Beta}(528, 924)$$

- For a 95% *credible* interval, $\alpha = 0.05$
 - Lower endpoint: qbeta(0.025, 528, 924)
 - Upper endpoint: qbeta(0.975, 528, 924)
 - $[\theta_{\alpha/2}, \theta_{1-\alpha/2}] = [0.34, 0.39]$

Interval for shooting skill in basketball

- Bayes credible interval: $[\theta_{\alpha/2}, \theta_{1-\alpha/2}] = [0.34, 0.39]$
- Frequentist *confidence* interval: $[0.39, 0.59]$
- End-of-season percentage was 0.37
- Credible intervals and confidence intervals have different meanings!

Highest Posterior Density (HPD) region

Definition: (HPD region) A $100 \times (1 - \alpha)$ HPD region consists of a subset of the parameter space, $R(y) \in \Theta$ such that

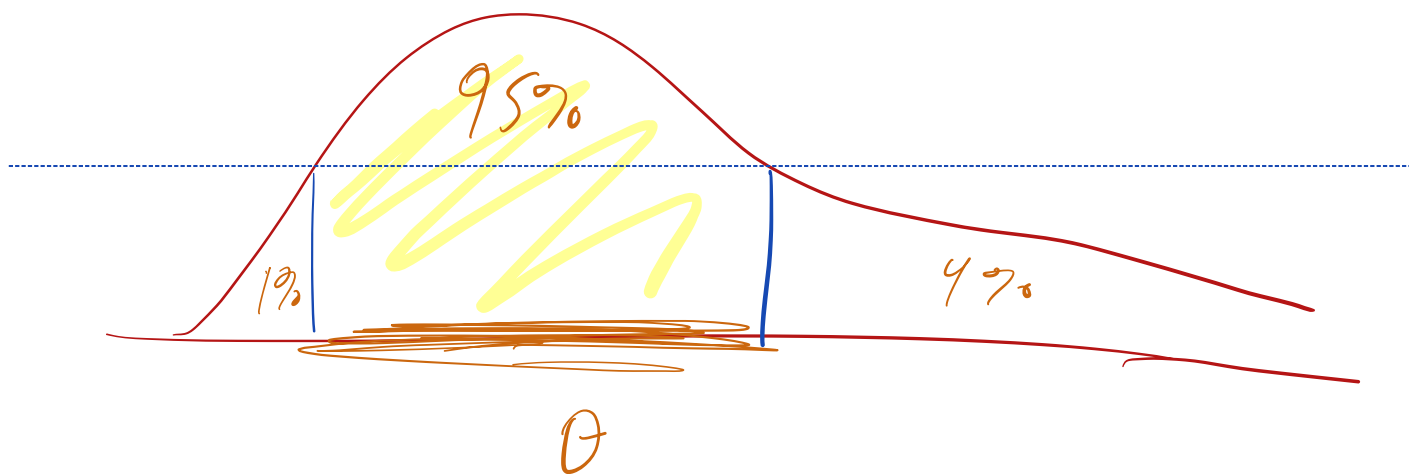
✓ 1. $\Pr(\theta \in R(y) | Y = y) = 1 - \alpha$ e.g. 95%

- The probability that θ is in the HPD region is $1 - \alpha$

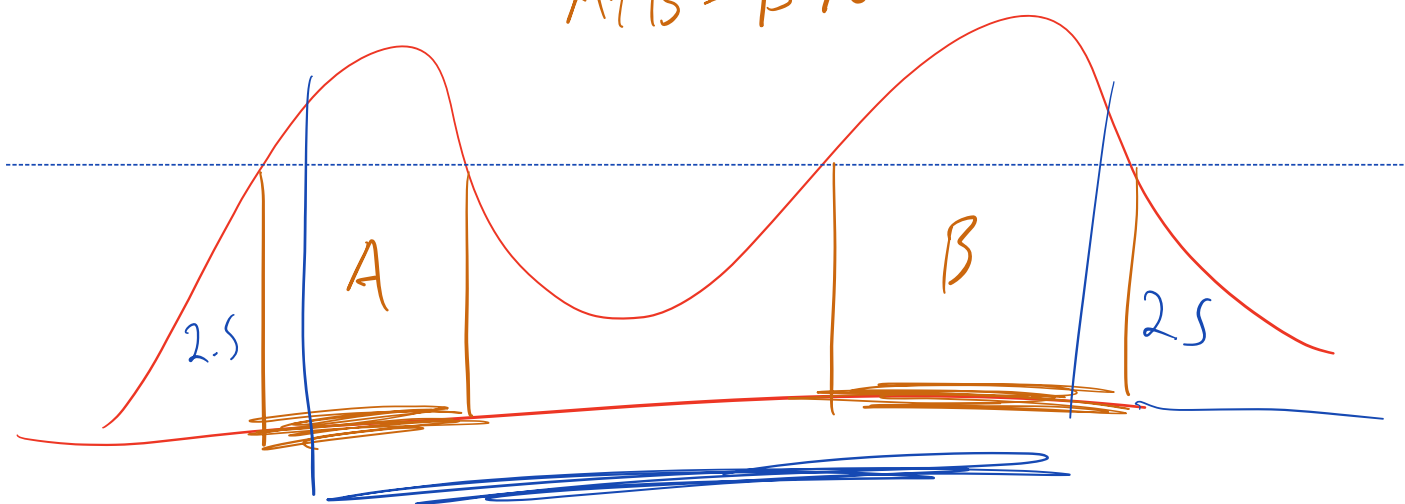
2. If $\theta_a \in R(y)$, and $\theta_b \notin R(y)$, then $p(\theta_a | Y = y) > p(\theta_b | Y = y)$

- All points in an HPD region have a higher posterior density than points outside the region.

The HPD region is the *smallest* region with probability $(1 - \alpha)$



$$A + B = 95\%$$



Frequentist behavior of Bayesian intervals

- Bayesian credible intervals usually won't have exactly correct frequentist coverage
- If our prior was well-calibrated and the sampling model was correct, we'd have well-calibrated credible intervals
- And: asymptotically, a central posterior interval will cover the true value 95% of the time under repeated sampling

$n \rightarrow \infty$

(well calibrated)

[Ch. 4, BDA]

$$\log P(\Theta|y) \approx \log P(\hat{\Theta}|y) + \frac{d \log P(\hat{\Theta}|y)}{d\Theta} (\Theta - \hat{\Theta}) + \frac{1}{2} (\Theta - \hat{\Theta}) \frac{d^2 \log P(\hat{\Theta}|y)}{d\Theta^2} (\Theta - \hat{\Theta}) + \text{higher order,}$$

(expand about $\hat{\Theta}$, posterior mode)

Bayes

$$P(\Theta|y) \xrightarrow[n \rightarrow \infty]{d} N(\hat{\Theta}, I(\hat{\Theta})^{-1})$$

$-I(\hat{\Theta})$

Posterior Mode.

Classical

$$\hat{\Theta}_{MLE} \approx N(\Theta, I(\Theta)^{-1})$$

Frequentist-Bayes Unification

Under regularity conditions:

$$p(\theta \mid y_1, \dots, y_n) \approx \mathcal{N} \left(\hat{\theta}, [I(\hat{\theta})]^{-1} \right)$$

Classical result:

$$\hat{\theta} \approx \mathcal{N}(\theta, [I(\theta)]^{-1})$$

Sequential Bayesian Updating

$$\begin{array}{ccccccc}
 \underbrace{p(\theta)}_{\text{prior}} & \rightarrow & \underbrace{p(\theta \mid y_1)}_{\text{reveal the first observation}} & \xrightarrow{\text{New Prior:}} & \underbrace{p(\theta \mid y_1, y_2)}_{\text{reveal the second observation}} & \dots \\
 & & & & & \\
 & \rightarrow \dots \rightarrow & \underbrace{p(\theta \mid y_1, \dots, y_n)}_{\text{reveal all } n \text{ observations}}
 \end{array}$$

When data are i.i.d., final posterior is the same, regardless of whether we analyze data sequentially or as a single batch.

$$P(\theta \mid y_1, \dots, y_n) \propto \underbrace{P(\theta) P(y_1 \mid \theta) P(y_2 \mid \theta) \dots P(y_n \mid \theta)}_{P(\theta \mid y_1, y_2, \dots, y_n)}$$

Improper prior distributions

- For the Beta distribution we chose a uniform prior (e.g. $p(\theta) \propto \text{const}$). This was ok because
 - $\int_0^1 p(\theta) d\theta = \text{const} < \infty$ $\theta \in [0, 1]$
 - We say this prior distribution is proper because it is integrable
- For the Poisson distribution, try the same thing:
 $p(\lambda) \propto \text{const}$
 - $\int_0^\infty p(\lambda) d\lambda = \infty$
 - In this case we say $p(\lambda)$ is an *improper* prior

Improper prior distributions

- Sometimes there is an absence of precise prior information
- The prior distribution does not have to be proper but the posterior does!
- A proper distribution is one with an integrable density
- If you use an improper prior distribution, you need to check that the posterior distribution is also proper

If using improper priors,
must check posterior is proper.

Objective Bayes

- Also called “default”, “reference”, “non-informative” prior distributions
- Laplace’s principle of insufficient reason
- Principle of maximum entropy (MAXENT).
- Matching prior distributions
 - Find prior distributions which lead to posterior intervals with approximate frequentist coverage
- Invariant priors

Choose uniform?

Entropy:

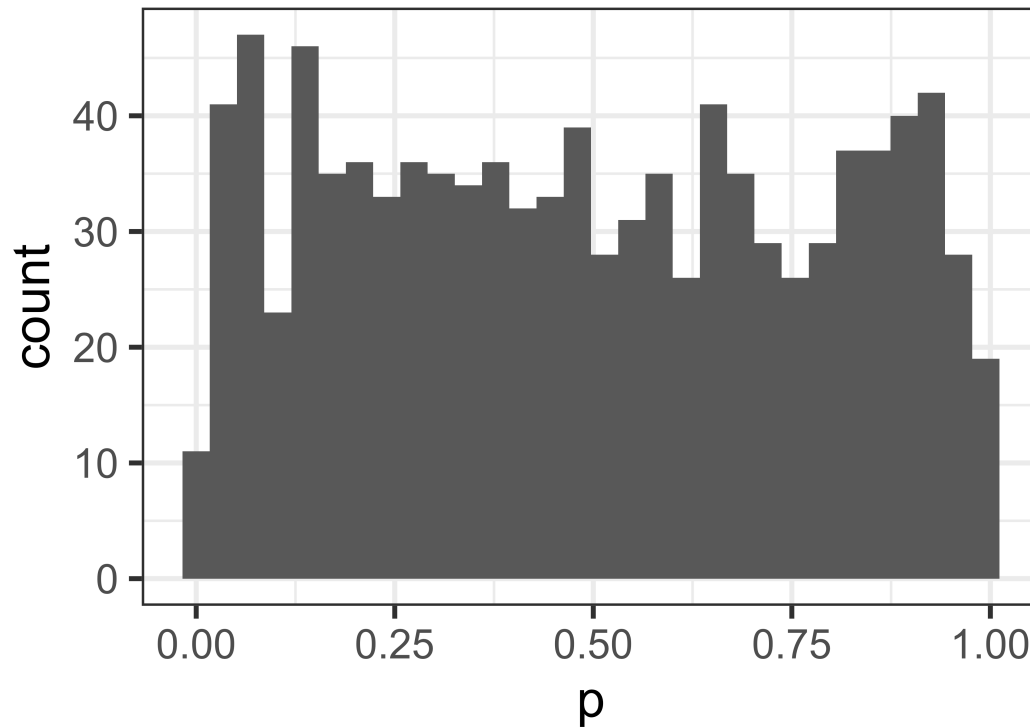
$$- \sum P(x) \log P(x)$$

For all distns
w/ fixed
mean & var,
 $N(\mu, \sigma^2)$

Laplace's principle of insufficient reason

Uniform distribution for p

```
1 p <- runif(1000)
2 tibble(p=p) %>% ggplot() +
3   geom_histogram(aes(x=p), bins=30) +
4   theme_bw(base_size=24)
```



Laplace's principle of insufficient reason

- Assume that $Y \sim \text{Bin}(n, \theta)$ but that we're most interested in the log odds:

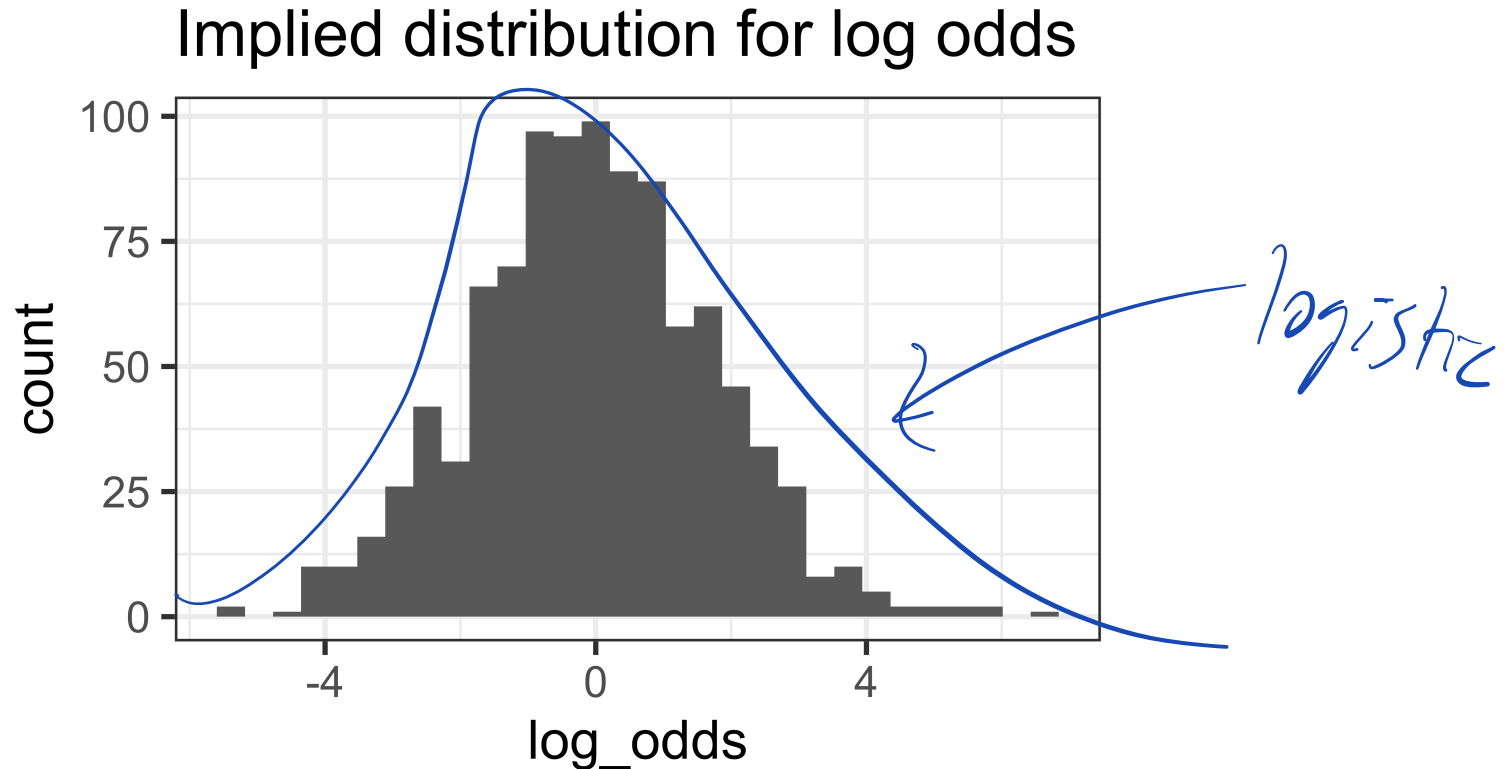
$$\gamma = \log \text{odds}(\theta) = \log \left(\frac{\theta}{1 - \theta} \right)$$

Handwritten note: "odds" written above the fraction in the equation.

- What prior should we use if we want to be “noninformative”?

Difficulties with non-informative priors

```
1 log_odds <- log(p/(1-p))  
2 tibble(log_odds=log_odds) %>% ggplot() +  
3   geom_histogram(aes(x=log_odds)) +  
4   theme_bw(base_size=24) +  
5   ggtitle("Implied distribution for log odds")
```



Method of transformations

Assume the prior density, $p(\theta)$. What is the implied prior density for the transformed parameter, $\gamma = g(\theta)$?

1. Find the inverse, $\theta = g^{-1}(\gamma)$

2. Compute $\frac{dg^{-1}(\gamma)}{d\gamma}$

3. Find $p_\gamma(\gamma) = \left| \frac{dg^{-1}(\gamma)}{d\gamma} \right| \times p_\theta(g^{-1}(\gamma))$

Jacobian.

$$\gamma = \log \frac{\theta}{1-\theta}$$

$$\theta = \frac{e^\gamma}{1+e^\gamma}$$

$$\frac{d e^\gamma}{d\gamma (1+e^\gamma)^2} = \frac{e^\gamma}{(1+e^\gamma)^2}$$

$$P(\gamma) \propto \frac{e^\gamma}{(1+e^\gamma)^2} \times p_\theta\left(\frac{e^\gamma}{1+e^\gamma}\right) = \frac{e^\gamma}{(1+e^\gamma)^2} \times \frac{1}{\sqrt{\pi}} \exp\left(-\frac{1}{2} \log^2 \frac{e^\gamma}{1+e^\gamma}\right)$$

Jeffreys prior

- Idea: using the geometry of the statistical model itself to define the prior
 - Specify a rule for constructing the prior that is invariant under transformations
- Derivation:

Jeffreys Prior: $P(\theta) \propto \sqrt{I(\theta)}$

$\phi = g(\theta)$, what is $P(\phi)$

$$P_\phi(\phi) = \left| \frac{d\theta}{d\phi} \right| P_\theta(\theta)$$

$$= \left| \frac{d\theta}{d\phi} \right| \sqrt{I(\theta)}$$

$$= \left| \frac{d\theta}{d\phi} \right| \sqrt{I(\phi) \left(\frac{d\phi}{d\theta} \right)^2}$$

$$= \sqrt{I(\phi)}$$

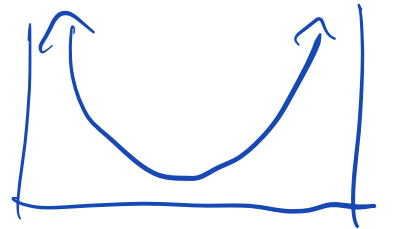
Prove

$$\begin{aligned} I_\phi(\phi) &= I_\theta(\theta) \left(\frac{d\theta}{d\phi} \right)^2 \\ I_\theta(\theta) &= I_\phi(\phi) \left(\frac{d\phi}{d\theta} \right)^2 \end{aligned}$$

$$Y \sim \text{Bin}(n, \theta)$$

$$P(\theta) \propto \sqrt{I(\theta)} \propto \theta^{-1/2} (1-\theta)^{-1/2}$$

$$\text{Beta}(1/2, 1/2)$$



$$\text{Unif equiv. Beta}(1, 1)$$

$$y \stackrel{\text{iid}}{\sim} N(\mu, \sigma^2)$$

known.

$$I(\mu) \propto \frac{n}{\sigma^2} \propto \text{const}$$

$$P(\mu) \propto \sqrt{I(\mu)} \propto \text{const.}$$

Improper Flat prior.

Leads to a proper prior.

Weakly Informative Priors

- A proper prior distribution, but intentionally include less information than is actually available a priori
- Construction:
 - Start with a noninformative prior and add then add enough information to constrain inferences to be more reasonable
 - Or: start with an informative prior and remove information
- Example: coefficients in a logistic regression should have a magnitude less than 10, in general

Prediction

Posterior predictive distribution

- An important feature of Bayesian inference is the existence of a predictive distribution for new observations.
 - Let $\tilde{y}$ be a new (unseen) observation, and $y_1, \dots, y_n$ the observed data.
 - The Posterior predictive distribution is $p(\tilde{y} \mid y_1, \dots, y_n)$

$$p(\tilde{y} \mid y_1, \dots, y_n) = \int p(\tilde{y} \mid \theta) p(\theta \mid y_1, \dots, y_n) d\theta$$

Posterior predictive distribution

- The predictive distribution does not depend on unknown parameters
- The predictive distribution only depends on observed data
- Asks: what is the probability distribution for new data given observations of old data?

Another Basketball Example

- I take free throw shots and make 1 out of 2. How many do you think I will make if I take 10 more?
- If my true “skill” was 50%, then ~~$\tilde{Y} \sim \text{Bin}(10, 0.50)$~~
- ~~Is this the correct way to calculate the predictive distribution?~~

No. $\theta \neq .5$

Posterior Prediction

If you know θ , then we know the distribution over future attempts:

$$\tilde{Y} \sim \text{Bin}(10, \theta)$$

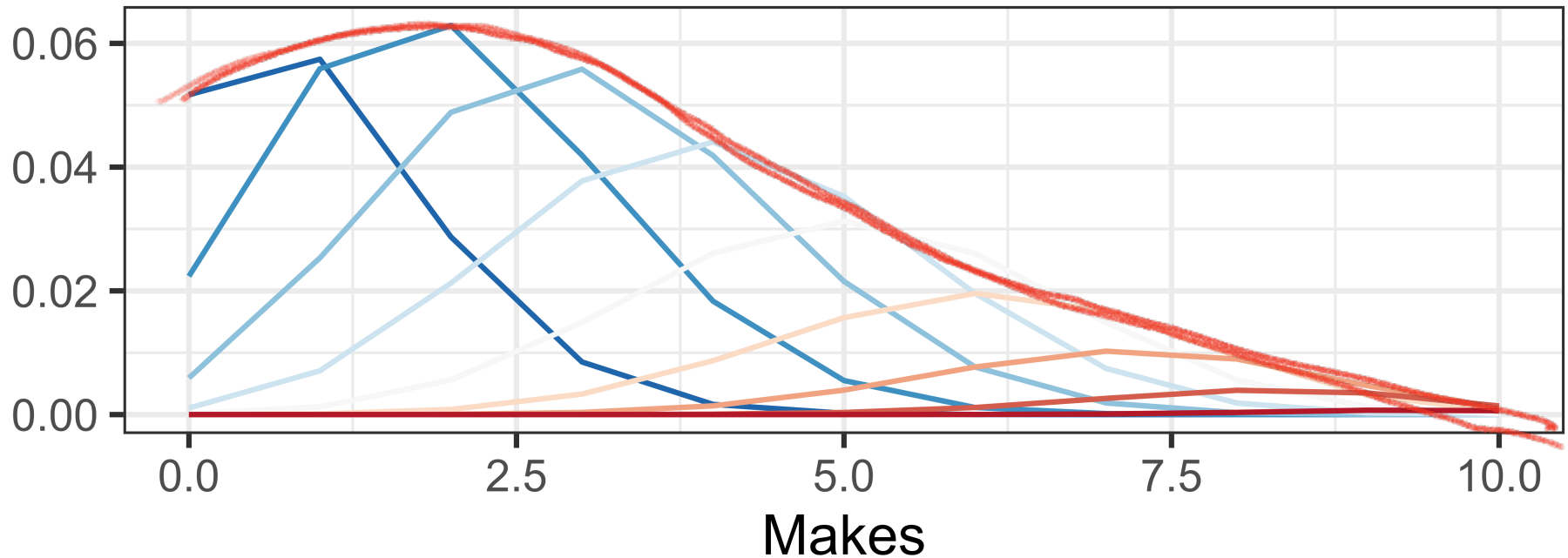
Posterior Prediction

- We already observed 1 make out of 2 tries.
- Assume a Beta(1, 3) prior distribution
 - e.g. a priori you think I'm more likely to make 25% of my shots
- Then $p(\theta \mid Y = 1, n = 2)$ is a Beta(2, 4)
- Intuition: weight $\tilde{Y} \sim \text{Bin}(10, \theta)$ by $p(\theta \mid Y = 1, n = 2)$

$$E[\theta] = \frac{\alpha}{\alpha + \beta} = 0.25$$

Posterior Prediction

If I take 10 more shots how many will I make?



theta

— 0.1	— 0.3	— 0.5	— 0.7	— 0.9
— 0.2	— 0.4	— 0.6	— 0.8	

$$P(\tilde{y} | y=1, n=2) = \int_0^1 P(\tilde{y} | \theta) P(\theta | y) d\theta$$

$$= \int_0^1 \underbrace{\binom{10}{\tilde{y}} \theta^{\tilde{y}} (1-\theta)^{10-\tilde{y}}}_{P(\tilde{y} | \theta)} \underbrace{\theta^{2-1} (1-\theta)^{4-1} \frac{\Gamma(6)}{\Gamma(2)\Gamma(4)}}_{\text{Beta}(2, 4)} d\theta$$

$$= \binom{10}{\tilde{y}} \frac{\Gamma(6)}{\Gamma(4)} \int_0^1 \theta^{\tilde{y}+1} (1-\theta)^{13-\tilde{y}} d\theta$$

$$= \binom{10}{\tilde{y}} \frac{\Gamma(6)}{\Gamma(4)} \frac{\Gamma(\tilde{y}+2) \Gamma(14-\tilde{y})}{\Gamma(16)}$$

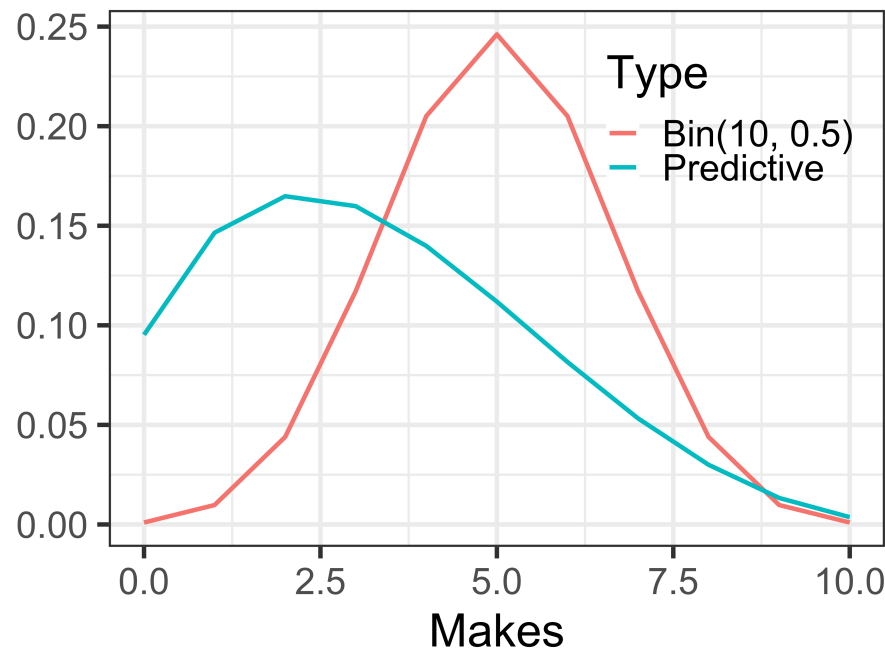
looks
like
a unnormalized
Beta.

Beta-Binomial.

Posterior predictive distribution

$$p(\theta) = \text{Beta}(1, 3), p(\theta | y) = \text{Beta}(2, 4)$$

$P(\tilde{y} | y)$
 ~~$P(\theta | y)$~~
 ~~$P(y | \theta_{\text{mtt}})$~~



The predictive density, $p(\tilde{y} | y)$, answers the question “if I take 10 more shots how many will I make, given that I already made

The posterior predictive distribution

$$\begin{aligned} p(\tilde{y} \mid y_1, \dots, y_n) &= \int p(\tilde{y}, \theta \mid y_1, \dots, y_n) d\theta \\ &= \int p(\tilde{y} \mid \theta) p(\theta \mid y_1, \dots, y_n) d\theta \end{aligned}$$

- The posterior predictive distribution describes our uncertainty about a new observation after seeing n observations
- It incorporates uncertainty due to the sampling in a model $p(\tilde{y} \mid \theta)$ and our posterior uncertainty about the data generating parameter, $p(\theta \mid y_1, \dots, y_n)$

The Beta-Binomial Distribution

law of iterated expectation

$$E[\tilde{y} | y] = E[E[\tilde{y} | y, \theta] | y]$$

$$= E[n \theta | y]$$

$$= n \frac{\alpha_{\text{post}}}{\alpha_{\text{post}} + \beta_{\text{post}}} = 10 \times \frac{2}{6} \approx 3.3$$

Law Total Variance. (EVE's)

$$\text{Var}(\tilde{y} | y) = \underline{E[\underline{\text{Var}}(\tilde{y} | y, \theta) | y]} + \text{Var}(E[\tilde{y} | y, \theta] | y)$$

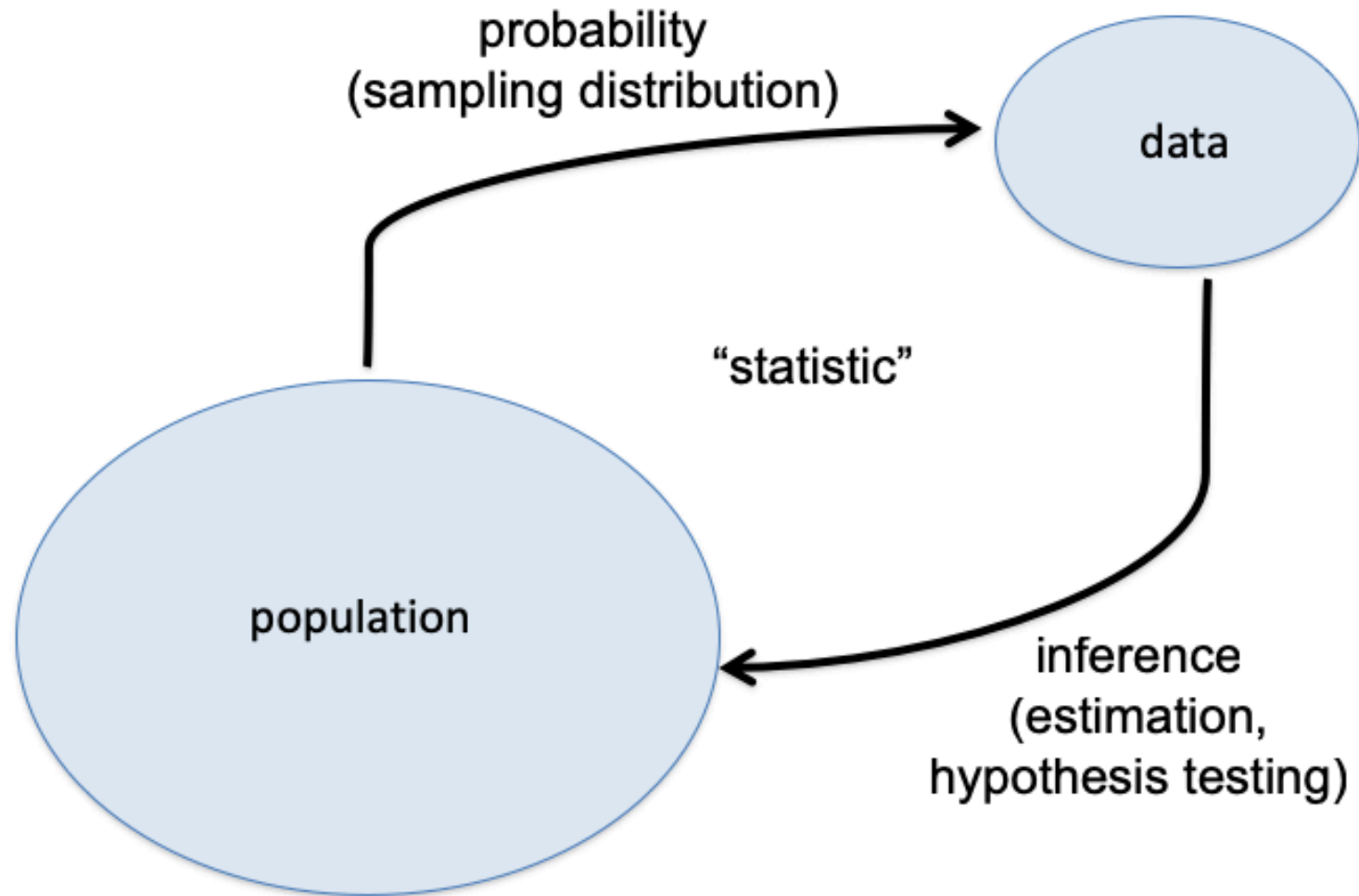
Posterior Predictive Variance.

$$\underline{E[n\theta(1-\theta) | y]} + \underline{\text{Var}(n\theta | y)}$$

Expected
Sampling Variance.

Variance due
Uncertainty About θ .

Posterior predictive density



The prior predictive distribution

$$\begin{aligned} p(\tilde{y}) &= \int p(\tilde{y}, \theta) d\theta \\ &= \int p(\tilde{y} \mid \theta) p(\theta) d\theta \end{aligned}$$

prior in
place
posterior

- The prior predictive distribution describes our uncertainty about a new observation before seeing data
- It incorporates uncertainty due to the sampling in a model $p(\tilde{y} \mid \theta)$ and our prior uncertainty about the data generating parameter, $p(\theta)$

Summary

- Conjugate priors
- Credible intervals
- Noninformative priors
- Posterior (or prior) predictive distribution